Pettit Student Edition 1m Yagi ERv1

Enhanced Reliability

This is the latest version of the 21cm Circular Patch Feed Disk Yagi

It combines the best of the **Original** all metal Yagi (aluminum spacers and components mounted on a threaded steel rod) with the much lighter version **Student Edition** (plastic spacers and components mounted on a wood dowel)

With identical performance to the Original, the ERv1 requires less components to be electrically connected thus increasing reliability It uses primarily the same lightweight Student Edition components to allow the antenna to be installed on inexpensive tripods.

B Alex Pettit Jr Apr2026

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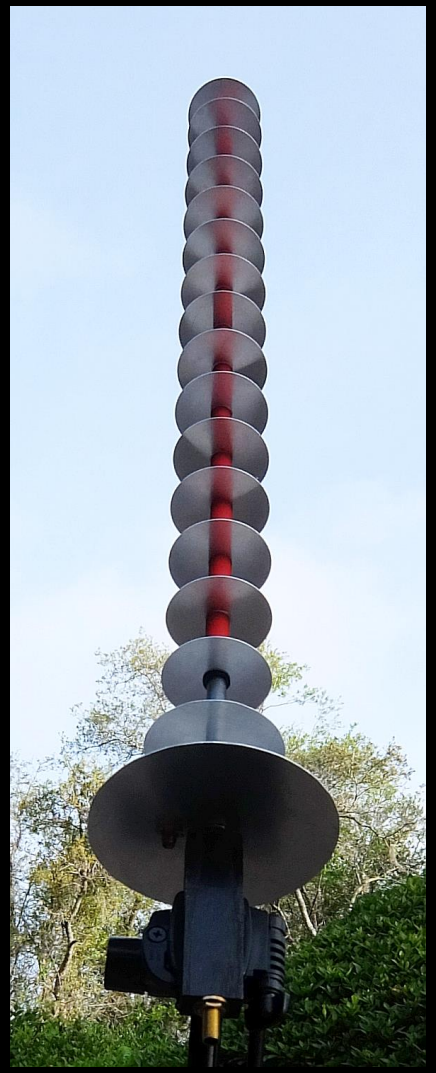

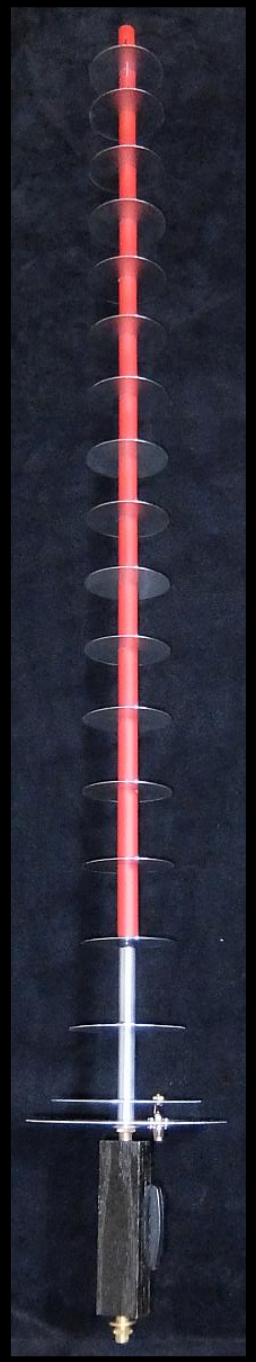

Results
&
Performance Comparison

Pettit Student Edition 1m Yagi ER Version 1

VS

Kraken Discovery Dish & H1 Feed

The Kraken Discovery Dish H1 System is the highest performing *Near-Off-The-Shelf* H Line Radio Telescopes I've evaluated. So it made a good system for reference

https://github.com/AP-HLine-3D/HLine3D/blob/main/21cm_Step1b_KrakenDiscoveryFeed_Eval_05.pdf

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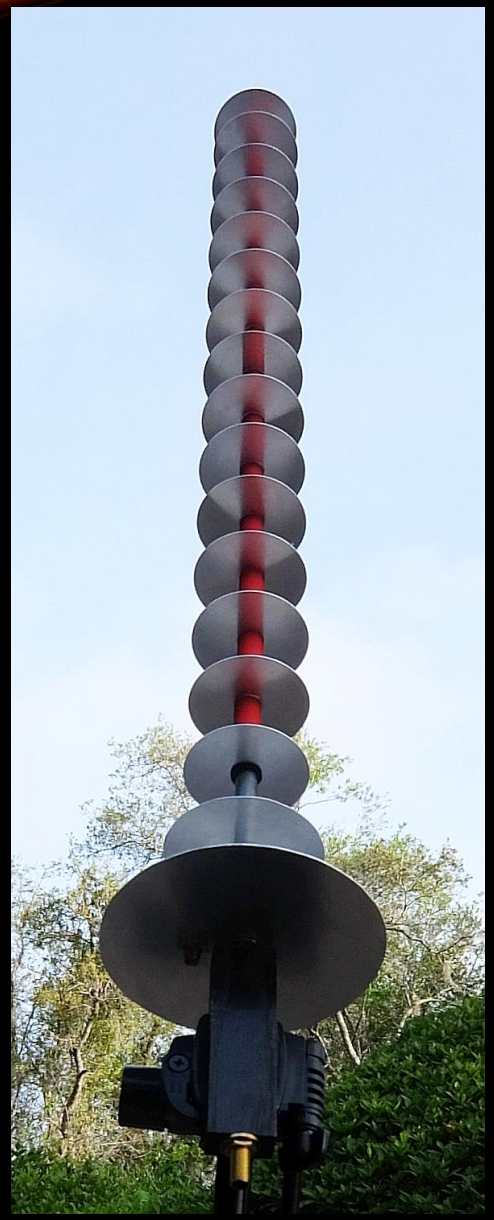

Kraken Discovery Feed 1420 MHz
w/ 1m x 0.66m Dish

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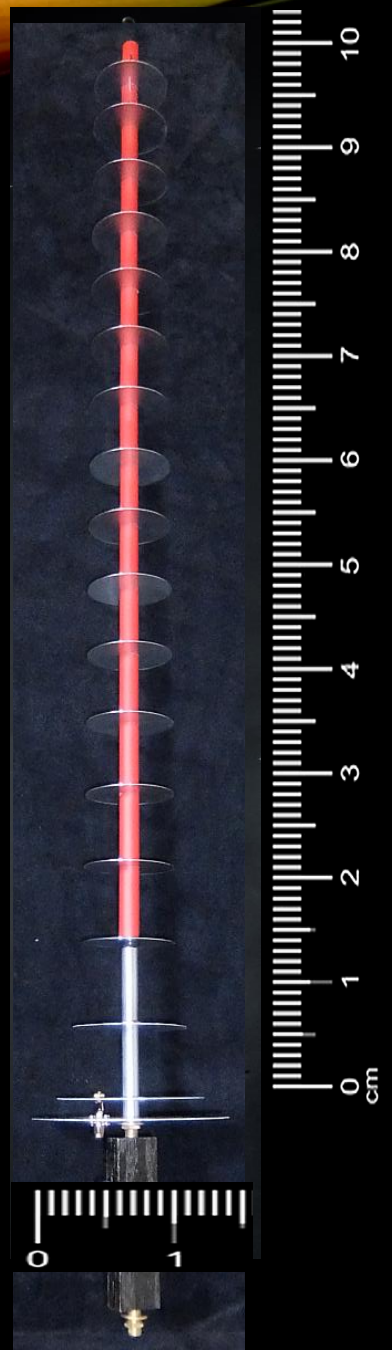

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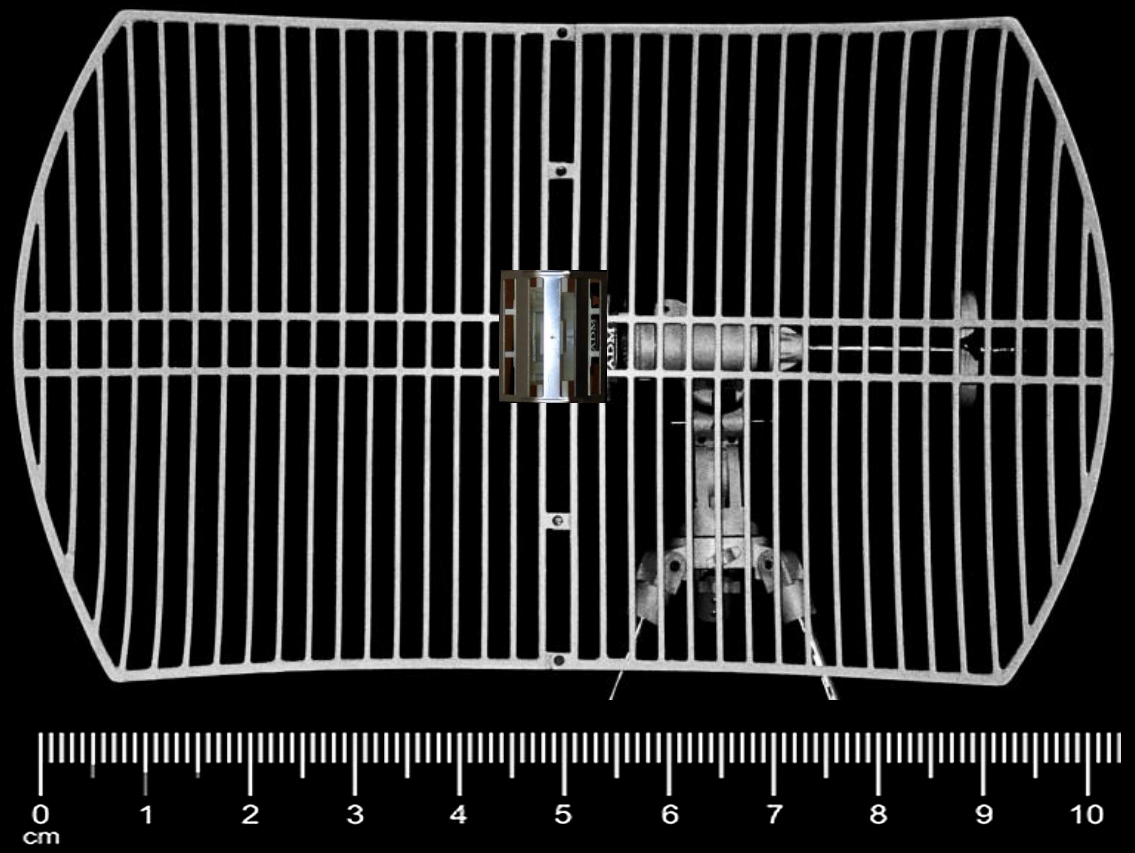

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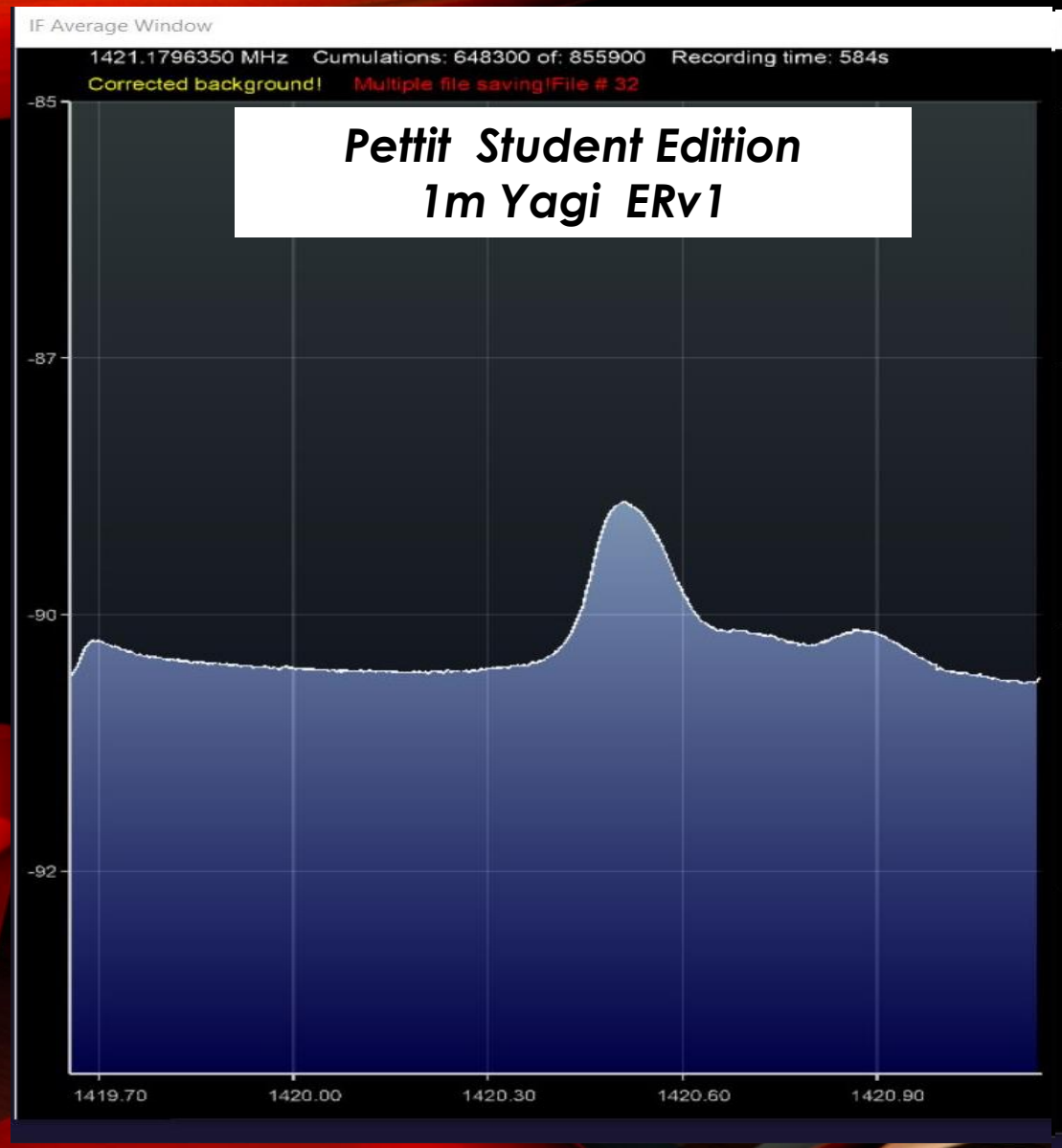
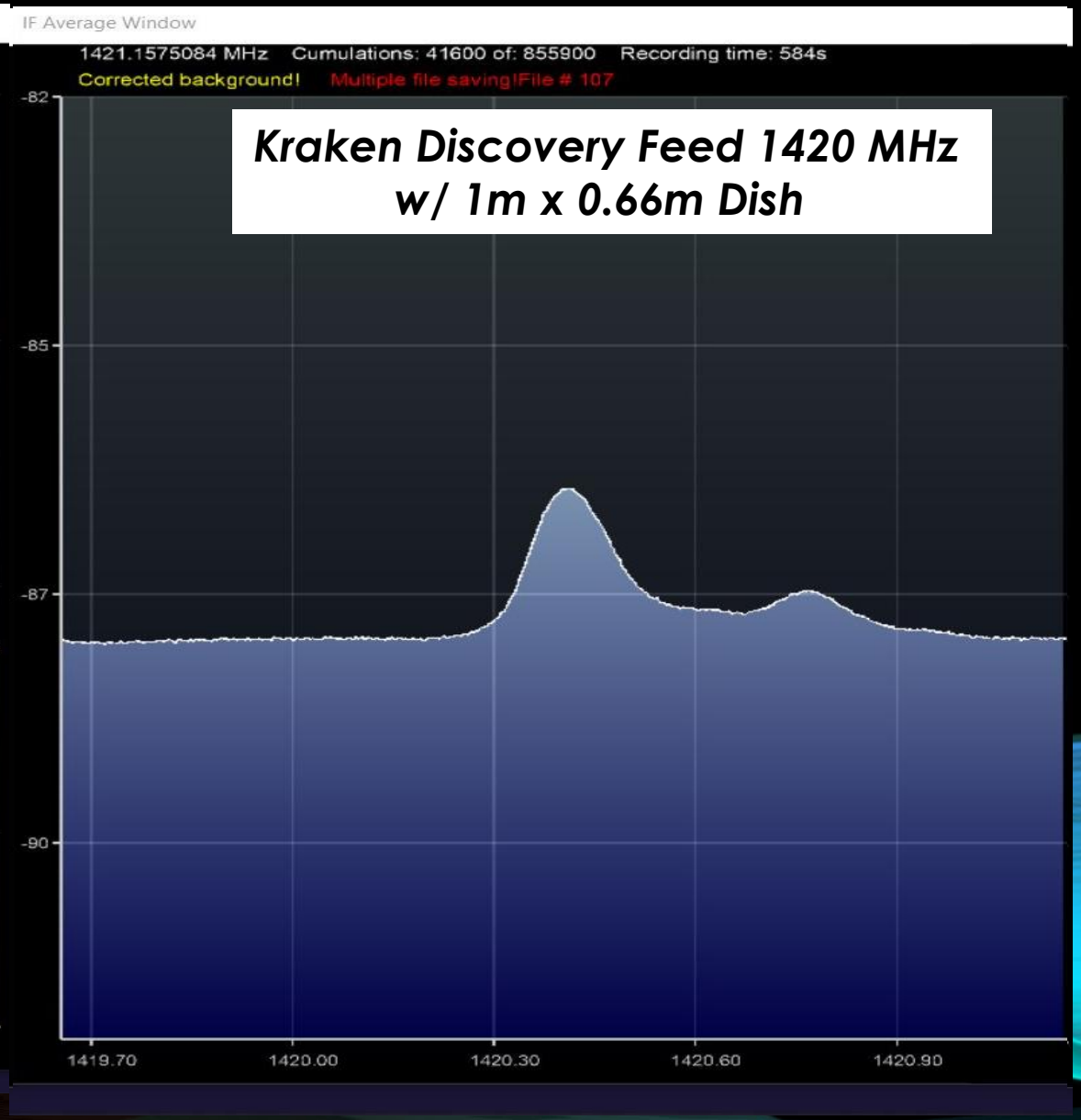
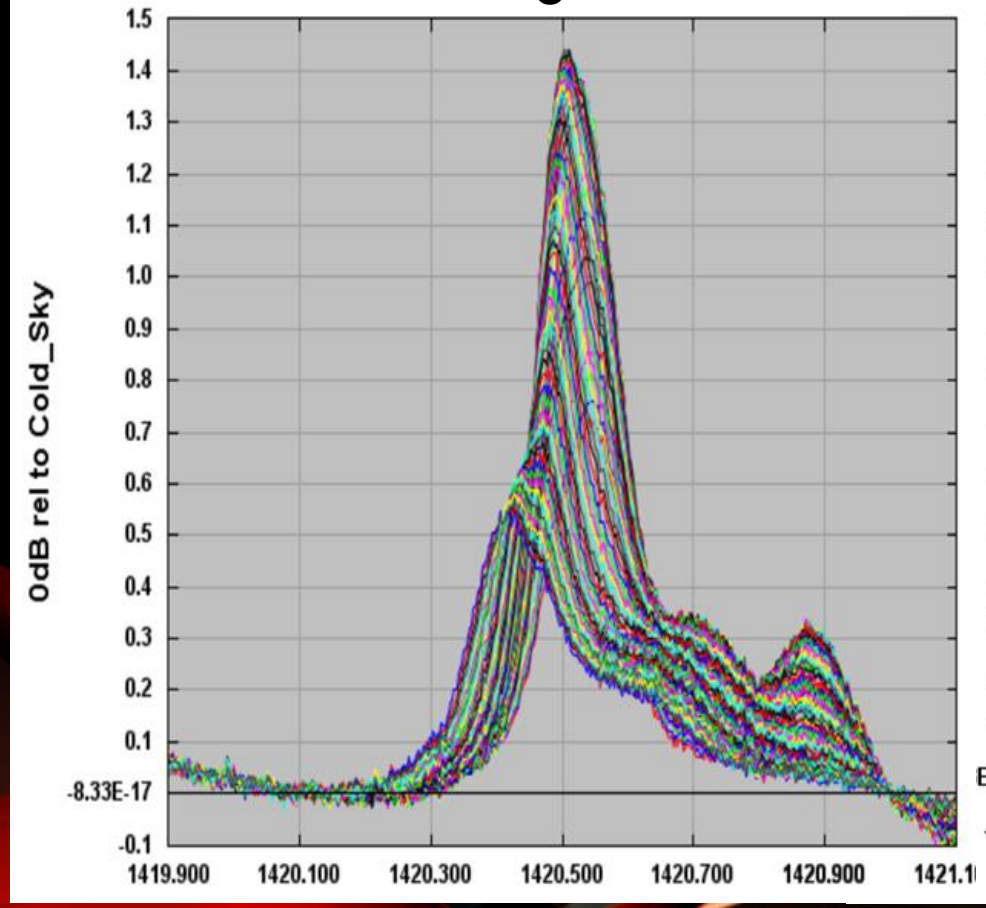

IF Average Window
1421.1796350 MHz Cumulations: 648300 of: 855900 Recording time: 584s
Corrected background! Multiple file saving!File # 32
-85
-87
-90
-92
1419.70
1420.00
1420.30
1420.60
1420.90
Pettit Student Edition
1m Yagi ERv1


IF Average Window
1421.1575084 MHz Cumulations: 41600 of: 855900 Recording time: 584s
Corrected background! Multiple file saving!File # 107
-82
-85
-87
-90
1419.70
1420.00
1420.30
1420.60
1420.90
Kraken Discovery Feed 1420 MHz
w/ 1m x 0.66m Dish

Due to its larger size, a 1m dia parabolic dish creates a beamwidth of 12 degrees vs 28 degrees for the smaller Yagi .
It improves the Angular Resolution of the Hydrogen distribution
But the Signal Strengths are quite similar.

Pettit Student Edition
1m Yagi ERv1



Kraken Discovery Feed 1420 MHz
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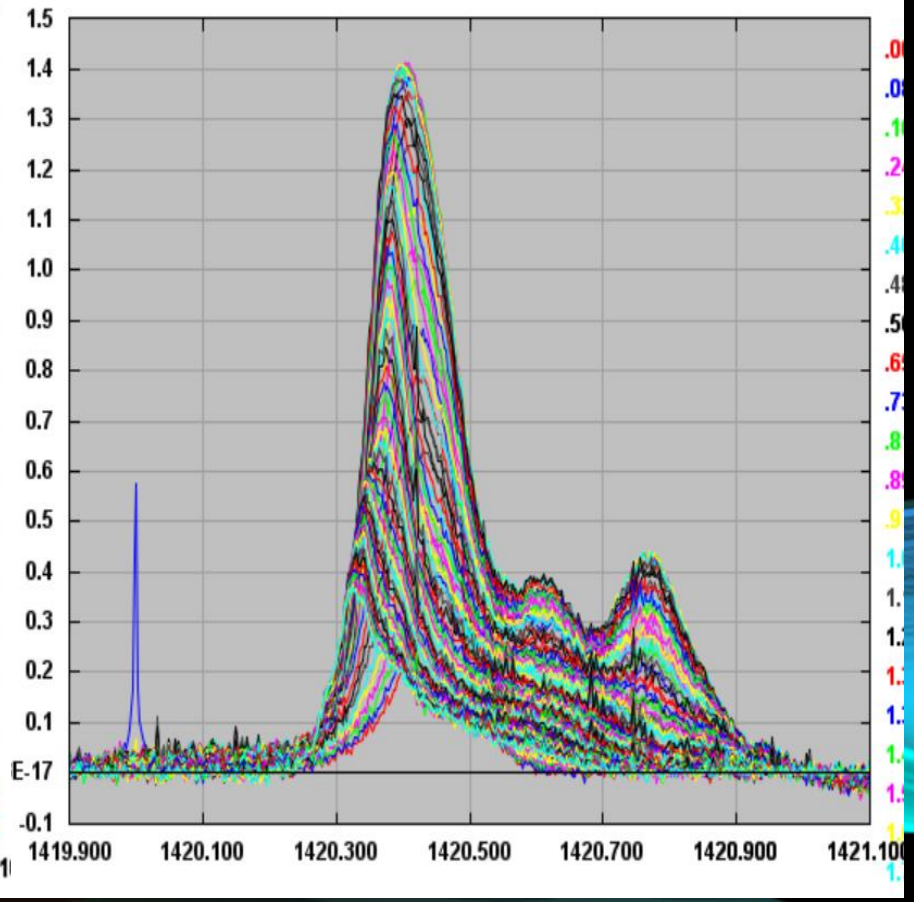

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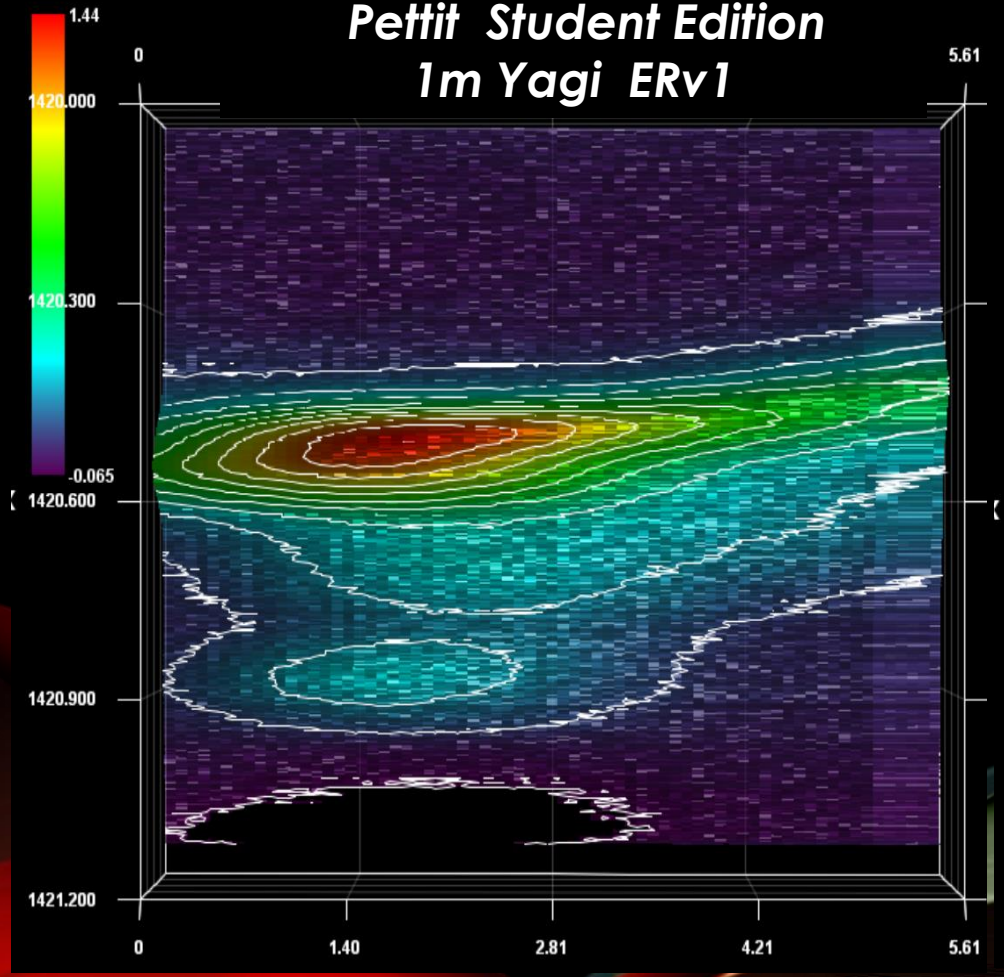

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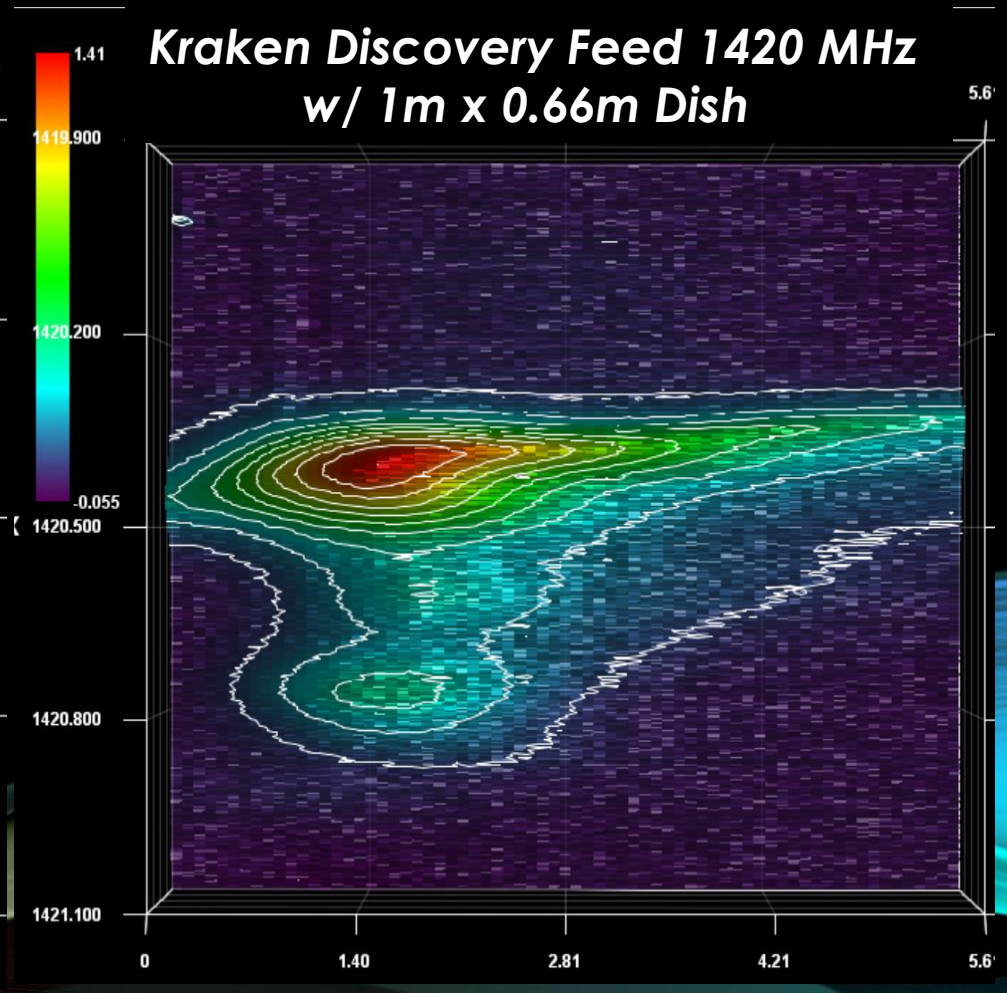

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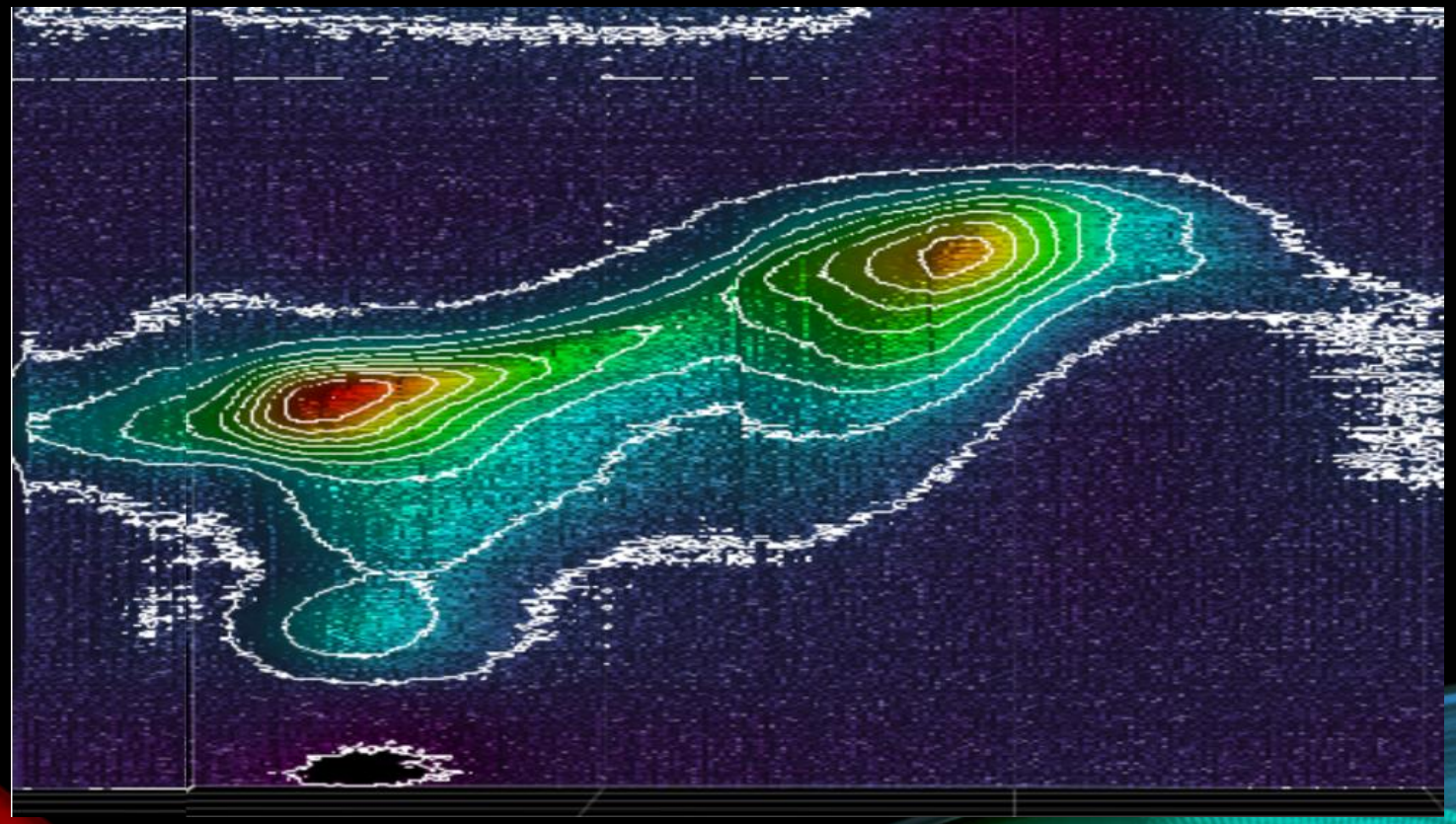
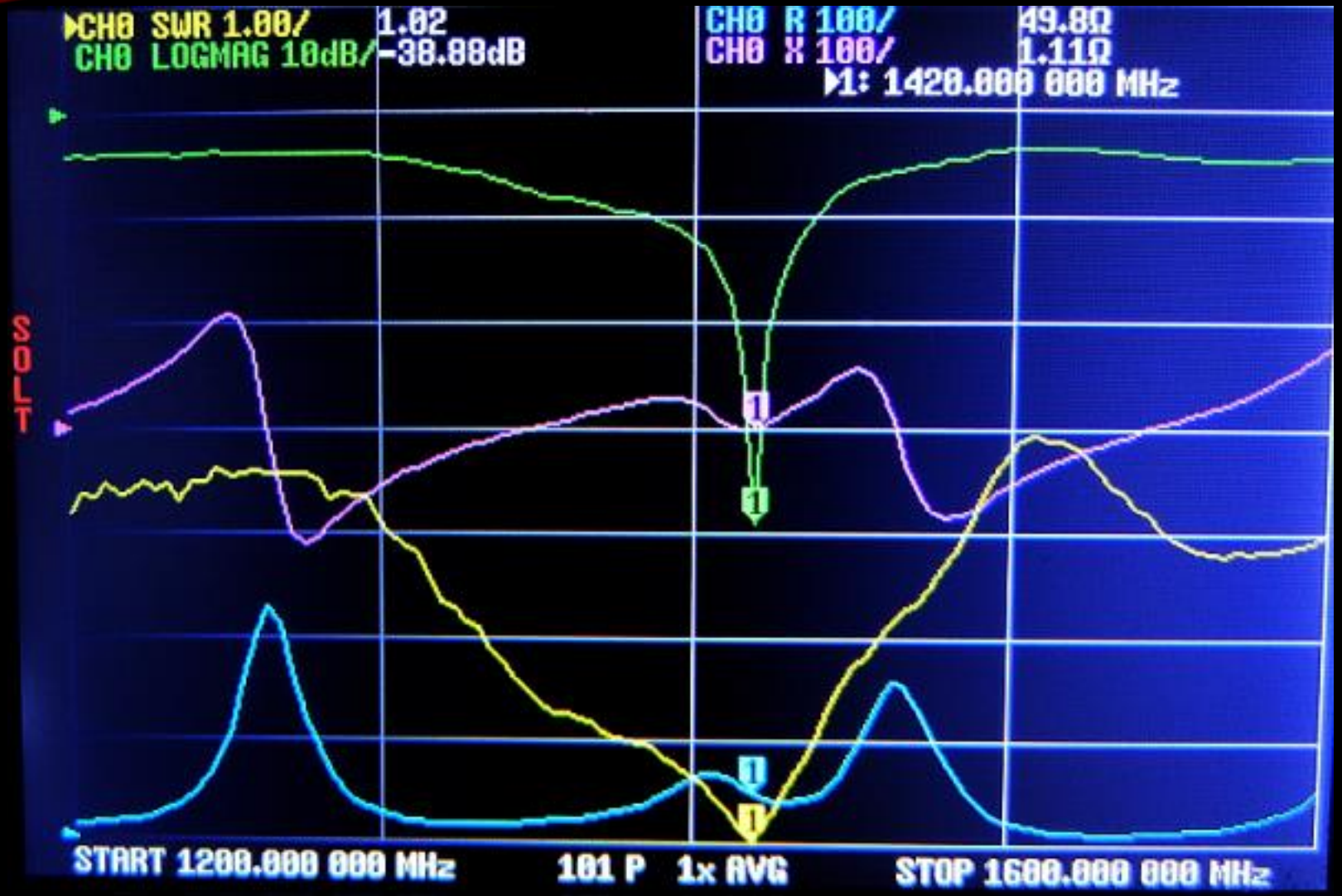
Antenna Tuning Performance